

Energy-efficient Analog Integrated Circuits

Sensor Interface – Exercise 3

Assignment

Gabriele Atzeni

Prof. Taekwang Jang

1. Consider the inverter-based amplifier with capacitive feedback from Exercise 2 - task 3.
 - a. Replace the feedback resistance R_F with the pseudo-resistor configurations seen in the lecture.
 - b. Discuss advantages and disadvantages of each implementation. Support your discussion using simulation results.

pseudo resistor should be 10GOhm

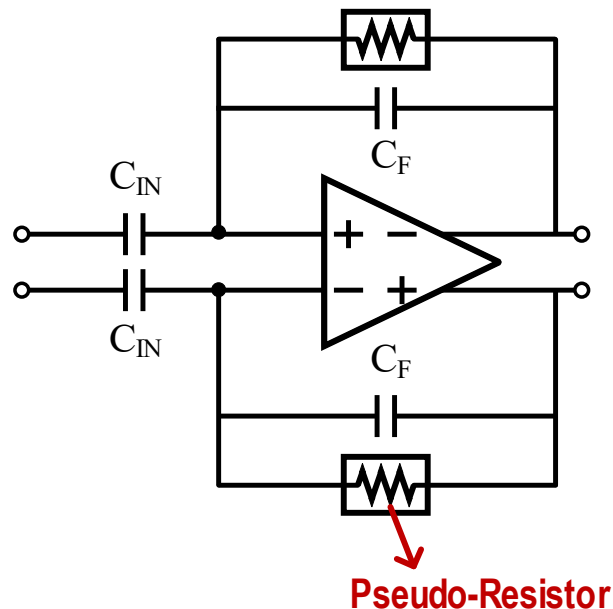


Fig. 1: Amplifier with capacitive feedback

2. Introduce chopping to eliminate low-frequency artifacts. Run periodic AC (pac), periodic noise (pnoise) and stability (pstb) analyses to simulate the transfer function, input-referred noise and noise efficiency factor of your chopper-stabilized amplifier. Assume a load capacitance of 100 fF and choose the chopping frequency considering that a signal bandwidth f_{BW} from 100 Hz to 5 kHz is required.

pss is needed in order to do other other simulations.
pac and pstb must be ran together

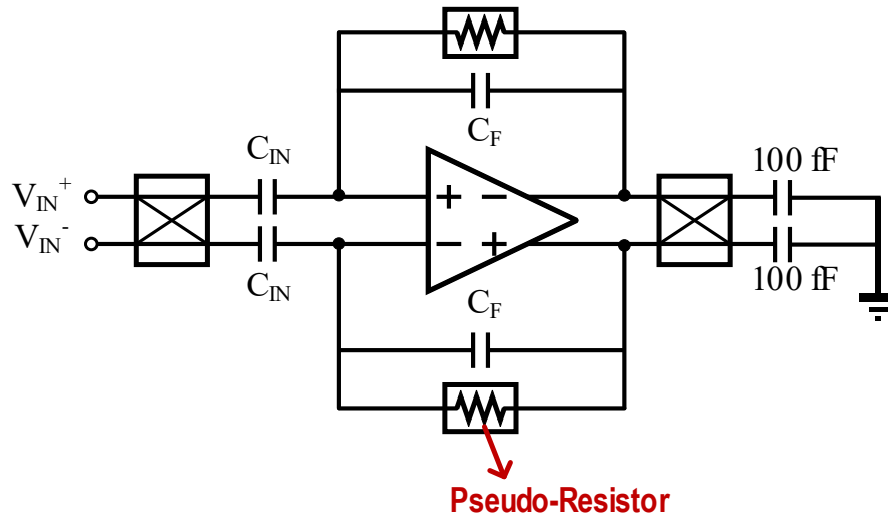


Fig 2: Chopper-stabilized amplifier

3. Consider the amplifier from Exercise 2 (question 4), where you introduced a mismatch in the width of the n-channel input pair.
 - a. Check the effect of chopping on the amplifier input-referred offset.
 - b. Sweep the duty cycle α of your chopping signals from 20% to 50% (Fig. 3). How does the duty cycle affect the residual offset?
4. Simulate the input-impedance of your chopper-stabilized amplifier using transient or pac analysis. Implement a capacitive positive feedback loop to boost the input impedance (Fig. 4).

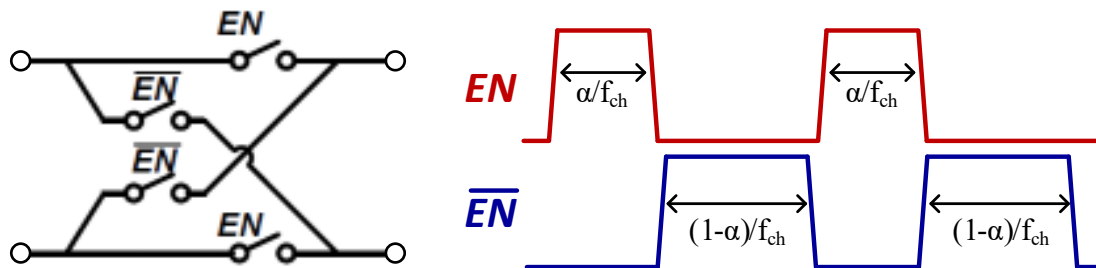


Fig 3. Chopping switches and waveforms

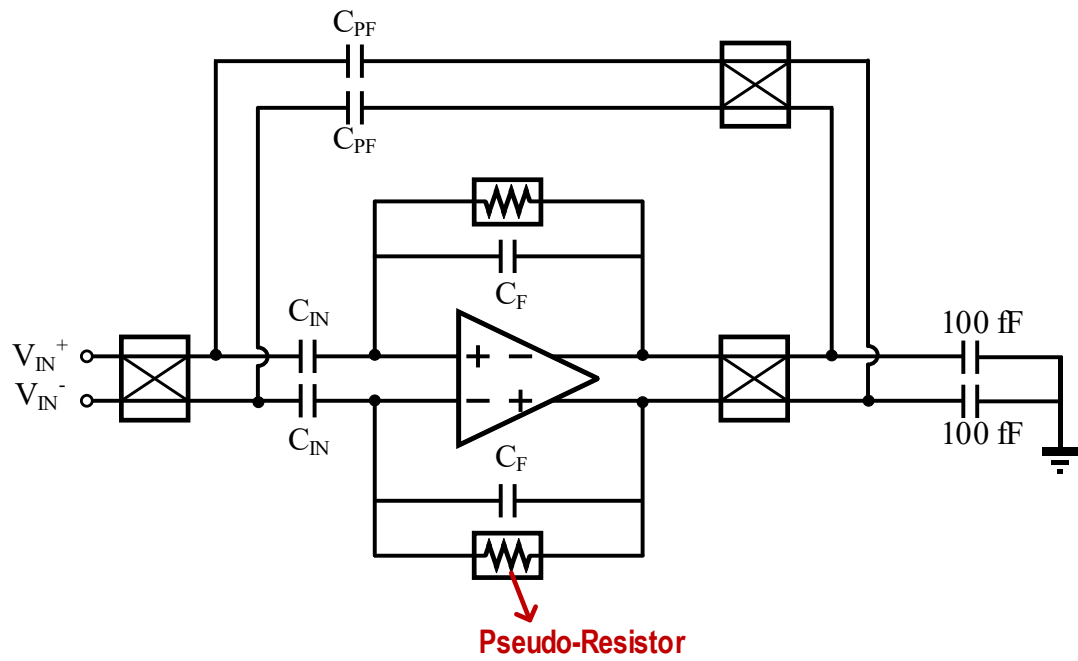


Fig 4: Chopper-stabilized amplifier with impedance-boosting loop